Solutions to JEE Main - 12 | JEE-2022

PHYSICS

SECTION-1

1.(C) We know that $E = \rho j$ (Ohm's Law)

Ratio of current densities, $$\frac{j_1}{j_2} = \frac{\left(\frac{i}{4\pi R_1^2}\right)}{\left(\frac{i}{4\pi R_2^2}\right)} = \left(\frac{R_2}{R_1}\right)^2 = \frac{25}{16}$$

Therefore, $$\frac{E_1}{E_2} = \frac{j_1}{j_2} = \frac{25}{16}$$

2.(A) We know that the resistance of a bulb is $R = \frac{V^2}{P}$

So, the 40 W bulb has a higher resistance, and therefore in series the potential difference across the 40 W bulb will be higher, and hence it will get damaged first as E is raised

Now, the resistance of the bulbs is

$$R_1 = \frac{200^2}{60} \quad \text{and} \quad R_2 = \frac{200^2}{40}$$

So, current through the bulbs is $i = \frac{E}{R_1 + R_2}$

So, potential difference across bulb B,

$$V_B = iR_2 = \frac{ER_2}{R_1 + R_2} = \left(\frac{\frac{1}{40}}{\frac{1}{60} + \frac{1}{40}}\right)E = 0.6E$$

So, for $V_B < 250$, E must be less than $\frac{250}{0.6}$, i.e. 416.7 V

3.(B) Radius of path, $r = \frac{mv}{qB}$

So, the velocity, $$v = \frac{qBr}{m}$$

Kinetic energy, $$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{qBr}{m}\right)^2 = \frac{q^2B^2r^2}{2m}$$

4.(C) In the balanced condition, i.e. when the current through the central branch is zero,

$$\frac{R_1}{R_2} = \frac{R_{AJ}}{R_{JB}} = \frac{25}{100-25} = \frac{1}{3} \quad \ldots\text{(i)}$$

After a resistance $15\ \Omega$ is connected in parallel with R_2,

$$\frac{R_1}{\left(\frac{15R_2}{15+R_2}\right)} = \frac{R_{AJ}}{R_{JB}} = \frac{50}{100-50} = 1 \quad \ldots\text{(ii)}$$

Solving (i) and (ii) simultaneously, we get $R_1 = 10\ \Omega$ and $R_2 = 30\ \Omega$

5.(D) The rotating ring is equivalent to a ring carrying a current $I = \frac{Q\omega}{2\pi}$

We know that the field at a point on the axis of a current carrying ring is

$$B = \frac{\mu_0 I R^2}{2\left(R^2 + x^2\right)^{3/2}}$$

So, the magnetic field intensity at the point $\left(0, 0, \sqrt{3}R\right)$ is

$$B = \frac{\mu_0 I R^2}{2\left(R^2 + \left(\sqrt{3}R\right)^2\right)^{3/2}} = \frac{\mu_0 I}{16R} = \frac{\mu_0 Q\omega}{32\pi R}$$

6.(B) At steady state, let the electric field in the vacuum between the plates be E

Then, the field inside the slab is $\frac{E}{\kappa} = \frac{E}{4}$

Therefore, $E\left(\frac{2d}{3}\right) + \left(\frac{E}{4}\right)\left(\frac{d}{3}\right) = V \quad \Rightarrow \quad E = \frac{4V}{3}$

So, potential difference across the slab,

$$V_{slab} = \left(\frac{E}{4}\right)\left(\frac{d}{3}\right) = \frac{V}{9}$$

7.(A) Using the right hand rule, the magnetic moment vector of the loop points in the –Z direction

Hence, the magnetic moment is $\vec{m} = Ia^2\left(-\hat{k}\right)$

8.(D) Heat loss $= \frac{1}{2}\frac{C_1 C_2}{C_1 + C_2}(V_1 - V_2)^2$

$$= \frac{1}{2}\frac{C(2C)}{C + 2C}(V_0 - 0)^2$$

$$= \frac{1}{3}CV_0^2$$

9.(B) Let the total resistance of the length of the wire that forms the loop be R

Then, in the first case, two resistances $\frac{R}{2}$ each are in parallel, and hence the net resistance is $\frac{R}{4}$

So, if the battery has emf E, then $I_0 = \frac{E}{\left(\frac{R}{4}\right)} = \frac{4E}{R}$

In the second case, two resistances, $\frac{R}{6}$ and $\frac{5R}{6}$ are in parallel, and hence the net resistance is $\frac{5R}{36}$

So, the current in the second case is $\frac{E}{\left(\frac{5R}{36}\right)} = \frac{36E}{5R} = \frac{9}{5}I_0$

10.(A) We know that if a charged particle is moving in a region with a uniform magnetic field, then the angle between its velocity and the magnetic field remains constant. If this angle is θ, then the path of the particle is:

(a) a straight line if $\theta = 0^o$ or $\theta = 180^o$

(b) a circle if $\theta = 90^o$

(c) a helix otherwise

In all cases, since the magnetic force on the particle is always perpendicular to its velocity, the magnitude of its velocity (and hence its kinetic energy) remains unchanged

11.(C) The upper branch:

(a) $2\ \mu F$ and $4\ \mu F$ in parallel $\Rightarrow$ $6\ \mu F$

(b) $6\ \mu F$ and $1\ \mu F$ in series $\Rightarrow$ $\frac{(6)(1)}{6+1} = \frac{6}{7}\ \mu F$

The lower branch:

(a) $4\ \mu F$ and $1\ \mu F$ in parallel $\Rightarrow$ $5\ \mu F$

(b) $5\ \mu F$ and $2\ \mu F$ in series $\Rightarrow$ $\frac{(5)(2)}{5+2} = \frac{10}{7}\ \mu F$

Finally, $\frac{6}{7}\ \mu F$ and $\frac{10}{7}\ \mu F$ in parallel $\Rightarrow$ $\frac{16}{7}\ \mu F$

12.(B) Equivalent emf of the batteries of emf E_1 and E_2

$$E = \frac{\frac{E_1}{r_1} + \frac{E_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}}$$

The current through the battery of emf E_3 is zero if

$$E = E_3 \quad \Rightarrow \quad E_1 r_2 + E_2 r_1 = E_3 (r_1 + r_2)$$

We can see clearly that a particular value of E_3 exists for which the above relation holds true. But the relation is independent of r_3

So, the current can be made zero by changing E_3 but not by changing r_3

13.(C) Capacitance with the slab introduced is $(4)(2) = 8\ \mu F$

Since the voltage source has been removed, the charge on the plates remains unchanged by the removal of the slab

This charge is $Q = (8)(100) = 800\ \mu C$

So, the potential energy stored with the slab inside is

$$U_1 = \frac{1}{2}\frac{Q^2}{(8 \times 10^{-6})} = \frac{1}{2}\frac{(800 \times 10^{-6})^2}{(8 \times 10^{-6})} = 0.04\text{ J}$$

And, the potential energy stored with the slab removed is

$$U_2 = \frac{1}{2}\frac{Q^2}{(2 \times 10^{-6})} = \frac{1}{2}\frac{(800 \times 10^{-6})^2}{(8 \times 10^{-6})} = 0.16\text{ J}$$

So, the work done in removing the slab is $W = U_2 - U_1 = 0.12\text{ J}$

14.(A) Since the centre lies is collinear with the straight portions, the magnetic field due to the straight portions is zero

Field due to the larger semi-circle,

$$B_1 = \frac{1}{2}\left(\frac{\mu_0 I}{2R}\right) = \frac{\mu_0 I}{4R}$$ (directed into the plane of the loop)

Field due to the smaller semi-circle,

$$B_2 = \frac{1}{2}\left(\frac{\mu_0 I}{2(R/2)}\right) = \frac{\mu_0 I}{2R}$$ (directed out of the plane of the loop)

So, by vector addition, the net field at the centre of the loop is

$$B_{NET} = B_2 - B_1 = \frac{\mu_0 I}{4R}$$ (directed out of the plane of the loop)

15.(C) Force on the two sides of the loop which are parallel to the wire cancel out.
On the two sides which are parallel to the wire attractive force on the closer side is more than repulsine force on the further side. So, net fore is attractive.

16.(C) At steady state (i.e. after a long time), the current through the branches containing the capacitors is zero, and current only flows through the loop containing the three resistances and the battery

This current is $i = \frac{100}{2+3+5} = 10 \text{ A}$

Therefore, the p.d. across the capacitors are respectively

$V_1 = (2+3)(10) = 50 \text{ V}$ and $V_2 = (3+5)(10) = 80 \text{ V}$

So, the charge on the capacitors is

$Q_1 = (5)(50) = 250\ \mu\text{C}$ and $Q_2 = (10)(80) = 800\ \mu\text{C}$

17.(D) Initial potential energy, $U_I = -(\vec{m}\cdot\vec{B}) = -mB\cos 30^\circ = -\frac{\sqrt{3}}{2}mB$

Final potential energy, $U_F = -(\vec{m}\cdot\vec{B}) = -mB\cos 150^\circ = \frac{\sqrt{3}}{2}mB$

Work done, $W = U_F - U_I = \sqrt{3}\,mB$

18.(C) We can deduce using the right hand rule that the magnetic field due to the infinite wire at a point on the positive Y-axis is along the –Z direction

Since the magnetic force on a moving charged particle is $\vec{F} = q(\vec{v}\times\vec{B})$, we can deduce that the force on the particle at this instant is along the direction $\left(\frac{-\hat{i}+\hat{j}}{\sqrt{2}}\right)\times(-\hat{k})$, i.e. $\left(\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right)$.

19.(A) During discharging, the charge on the capacitor as a function of time is

$Q(t) = CV\,e^{-t/RC}$

So, at $t = 2RC$, $Q = \frac{CV}{e^2}$

Therefore, the charge that has flown through the resistance until now is $CV - \frac{CV}{e^2} = \left(1 - \frac{1}{e^2}\right)CV$

20.(D) Since the particle is negatively charged, the electric force on it will be opposite to the electric field, i.e. the force will be vertically downward

Hence, the net downward force on the particle is

$F_{down} = mg + qE = (10^{-3})(10) + (2\times 10^{-4})(200) = 0.05 \text{ N}$

The magnetic force must balance this downward force, so

$$F_{mag} = qvB = 0.05 \qquad \Rightarrow \qquad B = \frac{0.05}{\left(2\times10^{-4}\right)(500)} = 0.5 \text{ T}$$

And we know that the magnetic force must be vertically upwards, i.e. along the +Z direction. Therefore, since we know that the magnetic force is $q\left(\vec{v}\times\vec{B}\right)$, and the particle is negatively charged, we can deduce using the right hand rule that the field must be along the –Y direction (Since $-\left(\hat{i}\times\left(-\hat{j}\right)\right)=\hat{k}$)

SECTION-2

21.(20) EMF of the cell = (Current through potentiometer wire) × (Resistance of balancing length)

Therefore, $$\varepsilon = (2)\left(\left(\frac{30}{120}\right)(40)\right) = 20 \text{ V}$$

22.(4.50)

If a combination of capacitors with equivalent capacitance C_{eq} is charged completely using a battery of emf E, the heat dissipated is the work done by the battery minus the increase in potential energy of the capacitor combination, i.e. $\left(C_{eq}E\right)E - \frac{1}{2}C_{eq}E^2 = \frac{1}{2}C_{eq}E^2$

In parallel, $C_{eq} = 1+2 = 3\ \mu\text{F}$

Therefore, $H_1 = \frac{1}{2}(3)E^2 = \frac{3}{2}E^2$

In series, $C_{eq} = \frac{(1)(2)}{1+2} = \frac{2}{3}\ \mu\text{F}$

Therefore, $H_2 = \frac{1}{2}\left(\frac{2}{3}\right)E^2 = \frac{1}{3}E^2$

So, $\frac{H_1}{H_2} = \frac{9}{2}$

23.(90) Total resistance of the potentiometer wire,

$$R = (0.1)(60) + (0.05)(60) = 9\ \Omega$$

Therefore, in the balanced condition, the current through the main circuit is

$$i = \frac{24}{9+3} = 2 \text{ A}$$

Potential difference across PS,

$$V_{PS} = (2)(0.1)(60) = 12 \text{ V}$$

Since the emf of cell E_2 is greater than the potential difference across PS, the balancing point will be found to the right of S

So, in the balanced condition, let the length SJ = x (in cm)

Then, as the emf of cell E_2 must be equal to the potential difference across PJ, we get

$$12 + (2)\left((0.05)x\right) = 15 \qquad \Rightarrow \qquad x = 30 \text{ cm}$$

So, in the balanced condition,

$$PJ = PS + SQ = 60 + 30 = 90 \text{ cm}$$

24.(8) Current density in the wire is $j = \dfrac{I}{\pi R^2}$

Taking a circular loop of radius $\dfrac{R}{4}$ on a cross-section of the conductor, with the centre of the loop on the axis of the conductor, and applying Ampere's Law,

$$B\left(2\pi\left(\frac{R}{4}\right)\right) = \mu_0\left(j\left(\pi\left(\frac{R}{4}\right)^2\right)\right)$$

$$\Rightarrow \qquad B = \frac{\mu_0 I}{8\pi R}$$

25.(2.50) At this instant, let the potential difference across the capacitor C_2 be V_2

Then, by conservation of charge,

$$5(100) = 5(50) + 10V_2 \qquad \Rightarrow \qquad V_2 = 25 \text{ V}$$

Now, applying KVL in the loop, the current through the resistance is

$$i = \frac{50-25}{10} = 2.5 \text{ A}$$

Chemistry

SECTION-1

1.(D) $t_1 = t_2$

$\Rightarrow \quad \frac{1}{k_1}\ell n\frac{100}{50} = \frac{1}{k_2}\ell n\frac{100}{4}$

$\Rightarrow \quad \frac{k_2}{k_1} = \frac{\ell n 25}{\ell n 2} = 4.65$

2.(B) $k_{1(300K)} = \frac{0.693}{20}$; $k_{2(320)} = \frac{0.693}{5}$; $\ln\frac{k_{2(320)}}{k_{1(300)}} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

$E_a = \frac{2.303R\ T_1T_2}{(T_2 - T_1)}\log\frac{k_2}{k_1} = \frac{2.303\times 8.314}{20\times 1000}\times 300\times 320\log 4 \Rightarrow 55.14 kJ/mol$

3.(C) $t = \frac{2.303}{\lambda}\log\frac{n_u + n_{Pb}}{n_u}$ $\qquad n_v =$ no. of mole of U^{238}

$t = \frac{2.303}{\lambda}\ln\left(\frac{0.2}{0.1}\right)$ $\qquad n_{Pb} =$ no. of mole of Pb^{206}

$t = t_{1/2}$

4.(C) $A_1 e^{-Ea_1/RT} = A_2 e^{-Ea_2/RT}$

$\frac{A_2}{A_1} = e^{(Ea_2 - Ea_1)/RT}$

$10^2 = \text{Exp}\left\{\frac{600}{RT}\right\}$ $\qquad R = 2$ cal/K-mol

$2\ln 10 = \frac{600}{2T}$

$T = \left\{\frac{300}{2\times 2.303}\right\}K$

5.(B) Both are true statements and the correct explanation of first statement will be half life of I order reaction is independent of initial concentration of reactant.

6.(C) $t_{1/2} = \frac{0.693}{k} \Rightarrow \frac{0.693}{3465\times 10^{-6}} \Rightarrow 200 sec.$

After one half life, moles of A remaining $\Rightarrow 0.05$

And total no. of moles $\Rightarrow 0.05 + 0.15 \Rightarrow 0.2$

$V_f = \frac{n_f}{n_i}\times V_i \Rightarrow \frac{0.2}{0.1}\times 2 \Rightarrow 4$ litre

$\therefore$ Final concentration of $A = \frac{0.05}{4} \Rightarrow 0.0125M$

7.(A) $\dfrac{0.02\times\dfrac{0.01}{100}}{10^{-3}}=K\ 0.02^n$

$\dfrac{0.04\times\dfrac{0.01}{100}}{0.25\times10^{-3}}=K\ 0.04^n$

$2^n=8$

$n=3$

8.(B) $K_a=\dfrac{c\alpha^2}{1-\alpha}\Rightarrow 1.6\times10^{-5}=\dfrac{0.01\times\alpha^2}{1-\alpha}$

$\alpha=\sqrt{\dfrac{1.6\times10^{-5}}{0.01}}=\sqrt{1.6\times10^{-3}}=0.04$ $\quad;\alpha=\dfrac{\wedge_m}{\wedge_m^{\infty}};\ \wedge_m=0.04\times380\times10^{-4}$

$\wedge_m=\dfrac{K}{1000\times C}$

So, $K=0.04\times380\times10^{-4}\times1000\times0.01$

$K=1.52\times10^{-2}Sm^{-1}$

9.(B) Kohlrausch law is valid for strong as well as weak electrolyte.

10.(B) $E^o_{X^-/AgX/Ag}=E^o_{Ag^+/Ag}+\dfrac{0.059}{1}\log K_{sp}$

11.(A) $K_{sp}(Cd(OH)_2)=[Cd^{2+}][OH^-]^2$

$10^{-14}=[Cd^{2+}][10^{-4}]^2$

$[Cd^{2+}]=10^{-6}M$

$E_{Cd^{2+}|Cd}=E^0_{Cd^{2+}|Cd}-\dfrac{0.0591}{2}\log\dfrac{1}{[Cd^{2+}(aq)]}\Rightarrow-0.40-\dfrac{0.0591}{2}\log\dfrac{1}{10^{-6}}=-0.5773V$

12.(C) $w=Z\times q\times$ current efficiency

$Z=\dfrac{96.5}{0.9\times2\times96500}=5.55\times10^{-4}$

13.(A) On electrolysis of dilute H_2SO_4 with Cu electrodes;

At anode: $Cu(s)\longrightarrow Cu^{+2}(aq)+2e^-$

14.(C) During discharge of lead storage battery, density of solution decreases. During discharging,

Overall cell reaction is: $Pb+PbO_2+4H^++2SO_4^{2-}\longrightarrow2PbSO_4+2H_2O$

15.(C) $Au^++e^-\longrightarrow Au(s)\qquad E^\circ=1.69\ V\qquad ...(1)\qquad \Delta G_1^o$

$Au^{3+}+3e^-\longrightarrow Au(s)\qquad E^\circ=1.40\ V\qquad ...(2)\qquad \Delta G_2^o$

From (2) – (1)

$Au^{3+}+2e^-\longrightarrow Au^+$

$\Delta G_3^\circ=\Delta G_2^\circ-\Delta G_1^\circ$

$-2\times F\times E^\circ=-3\times F\times1.40+1\times1.69\times F$

$E^\circ=1.255\ V$

16.(D) The formation of micelle take place only above certain temperature called Kraft temperature suggests positive ΔS of micelle formation which even overcome effect of positive ΔH of micelle formation. Besides kinetic effect also become important at high temperature.

17.(D) Lyophobic sols are irreversible sols. Micelle formation takes place above Kraft temperature and PO_4^{3-} ion has less coagulation value than SO_4^{2-} as PO_4^{3-} ion has more negative charge so it has more coagulation power. No. of particles in colloidal sol is less so values of colligative properties will also be low.

18.(A) Activation energy is not required in physisorption.

19.(A) clay, eosin, charcoal are negatively charged sol and hemoglobin is positively charged sol.

20.(B) As_2S_3 is negatively charged sol

SECTION-2

21.(600)

$$\begin{array}{lcccc} & A & + \; B \longrightarrow & C & + \; D \\ t=0 & 1 & 1 & & \\ 't' & 1-x & 1-x & x & x \end{array}$$

rate $= k[A]^{\frac{1}{2}}[B]^{\frac{1}{2}}$

$$-\frac{d[A]}{dt} = k(1-x)^{1/2}(1-x)^{1/2} \qquad \Rightarrow -\frac{d(1-x)}{dt} = k(1-x) \;\; \Rightarrow \frac{dx}{dt} = k(1-x)$$

$$\Rightarrow \int_0^x \frac{dx}{1-x} = k\int_0^t dt \qquad \Rightarrow [-\ln(1-x)]_0^x = kt \qquad \Rightarrow -\ln(1-x) = kt$$

$$\Rightarrow -\ln(1-x) = kt \qquad \Rightarrow t = \frac{1}{k}\ln\left(\frac{1}{1-x}\right)$$

When $[A] = 0.25$

$\therefore \; x = 0.75$

$$\therefore \; t = \frac{1}{2.31\times10^{-3}}\ln\frac{1}{0.25} = \frac{10^3}{2.31}\ln 4 = \frac{2\times10^3\times0.693}{2.31} = \frac{2\times10^3\times0.693}{2.31} = 600 \text{ Sec}.$$

22.(1000) $\lambda = \dfrac{0.693}{t_{1/2}} = \dfrac{0.693}{69.3} = 0.01;$ $\quad \therefore \; N = \dfrac{-\frac{dN}{dt}}{\lambda};\; N = \dfrac{10}{10^{-2}} = 1000$ atoms

23.(3) log (23.5) = 1.37 & log (95) = 1.97

$$n = 1 + \frac{\log[(t_{1/2})_1/(t_{1/2})_2]}{\log(P_2/P_1)} \Rightarrow 1 + \frac{1.37-1.97}{\log(1/2)} = 3$$

24.(45) K = conductance × cell constant

$$K = \frac{1}{100}\times\frac{2}{4}$$

$$K = \frac{1}{200}\,\text{S cm}^{-1}$$

$$\wedge_m = \frac{k \times 1000}{C}$$

$$\wedge_m = \frac{1}{200} \times \frac{1000}{0.1}$$

$$\wedge_m = 50$$

$$\alpha = \frac{\wedge_m}{\wedge_m^{\infty}} = \frac{50}{400} = \frac{1}{8}$$

$$\underset{1-\alpha}{CH_3COOH} \rightleftharpoons \underset{\alpha}{CH_3COO^-} + \underset{\alpha}{H^{\oplus}}$$

$$i = 1 + \alpha = \frac{9}{8}$$

$$\Delta T_b = K_b \times m \times i$$

$$= 0.5 \times 0.1 \times \frac{9}{8} \times 800 = 45$$

25.(4) $k = \frac{\wedge_m . M}{1000} = \frac{200 \times 0.04}{1000} = 8 \times 10^{-3}\ Scm^{-1}$

$$k = G\left(\frac{\ell}{A}\right) \Rightarrow 8 \times 10^{-3} = G\left(\frac{8}{4}\right)$$

$$G = 4 \times 10^{-3}$$

$$V = IR = \frac{I}{G}$$

$$I = V \times G = 1000 \times 4 \times 10^{-3}\ A \Rightarrow 4A$$

Mathematics

SECTION-1

1.(B) $I=\int_0^{\frac{\pi}{2}}\frac{x\sin 2x}{\sin^4 x+\cos^4 x}dx$ and $I=\int_0^{\frac{\pi}{2}}\frac{\left(\frac{\pi}{2}-x\right)\sin 2x}{\cos^4 x+\sin^4 x}dx$

$$\therefore \quad 2I=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{\sin 2x}{1-\frac{1}{2}\left(1-\cos^2 2x\right)}dx=\frac{\pi}{2}\int_0^{\frac{\pi}{2}}\frac{\sin 2x}{\frac{1}{2}\left(1+\cos^2 2x\right)}dx$$

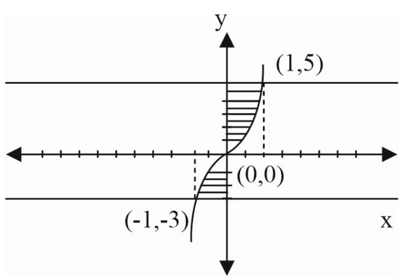
$$\therefore \quad I=\frac{\pi^2}{8}$$

2.(C) $L=\lim_{n\to\infty}\left(\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)......\left(1+\frac{n}{n}\right)\right)^{1/n}$

$$\therefore\ \ln L=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(1+\frac{r}{n}\right)=\int_0^1\ln(1+x)\,dx$$

$$\therefore\ \ln L=2\ln 2-1\Rightarrow L=\frac{4}{e}$$

3.(C)



$$=\int_{-1}^{0}\left(\left(x^3+3x+1\right)-(-3)\right)dx+\int_0^1\left(5-\left(x^3+3x+1\right)\right)dx=\frac{9}{2}$$

4.(A) Let $\int_0^x f(t)\,dt=T(x)\Rightarrow T'(x)=f(x)$

$\therefore$ On differentiating b.t.s. w. r.t. x, we get $f(x)=T'(x)$

Hence $G(x)=\int e^x\left(\int_0^x f(t)\,dt+f(x)\right)dx=\int e^x\left(T(x)+T'(x)\right)dx=e^xT(x)+C$

$$\Rightarrow G(x)=e^x\int_0^x f(t)\,dt+C$$

Now on differentiating $G'(x)=e^x\int_0^x f(t)\,dt+e^xf(x)\qquad\Rightarrow G'(0)=f(x)=1$

5.(C) $\sin x = t;\ I = \int \frac{\left(1-t^2\right)\left(2-t^2\right)}{t^2\left(1+t^2\right)}dt;$

$$f(t) = \frac{(y-1)(y-2)}{y(1+y)} = 1 + \frac{2(1-2y)}{y(y+1)};\ y = t^2 = 1 + 6\left[\frac{1}{3y} - \frac{1}{y+1}\right];\ \int\left(1 + \frac{2}{t^2} - \frac{6}{1+t^2}\right)dt$$

6.(A) $I = \int_1^\infty \frac{dx}{\left(e.e^x + e^3.e^{-x}\right)} = \int_1^\infty \frac{e^x dx}{e\left(e^{2x}+e^2\right)}$

(multiply Nr and Dr by e^x)

$$I = \frac{1}{e}\int_e^\infty \frac{dt}{t^2+e^2} = \frac{1}{e^2}\tan^{-1}\frac{t}{e}\Big|_e^\infty = \frac{1}{e^2}\left[\frac{\pi}{2} - \frac{\pi}{4}\right] = \frac{\pi}{4e^2}$$

7.(C) $T_r = \frac{1}{\sqrt{\frac{r}{n}}.n\left(3\sqrt{\frac{r}{n}}+4\right)^2}$

$$S = \frac{1}{n}\sum_{r=1}^{4n}\frac{1}{\left(3\sqrt{\frac{r}{n}}+4\right)^2.\sqrt{\frac{r}{n}}} = \int_0^4 \frac{dx}{\sqrt{x}\left(3\sqrt{x}+4\right)^2}$$

Put $3\sqrt{x}+4 = t \quad \Rightarrow \frac{3}{2}\frac{1}{\sqrt{x}}dx = dt = \frac{2}{3}\int_4^{10}\frac{dt}{t^2} = \frac{2}{3}\left[\frac{1}{t}\right]_{10}^4 = \frac{2}{3}\left[\frac{1}{4} - \frac{1}{10}\right] = \frac{2}{3}.\frac{6}{40} = \frac{1}{10}$

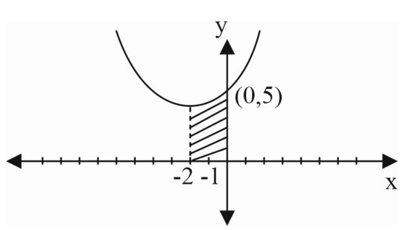
8.(D) $l = \ln\lim\limits_{t\to 0}\frac{\int_0^t (1+2\sin 3x)^{4/x}dx}{t} = \ln\lim\limits_{t\to 0}(1+2\sin 3t)^{4/t}$

(using L 'Hospital's rule)

$$= \ln e^{\lim\limits_{t\to 0}\frac{4}{t}(2\sin 3t)} = \lim_{t\to 0}\frac{2.3.4\sin 3t}{3t} = 24$$

9.(B) $y = x^2 + 4x + 5 = (x+2)^2 + 1$

$$A = \int_{-2}^0 \left(x^2+4x+5\right)dx = \frac{x^3}{3} + 2x^2 + 5x\Big]_{-2}^0$$



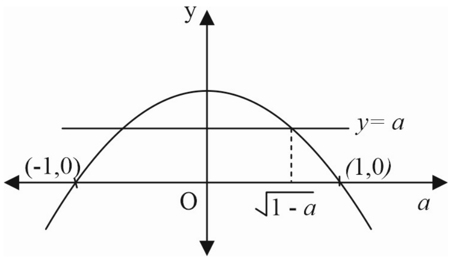
$$= -\left[-\frac{8}{3} + 8 - 10\right] = 2 + \frac{8}{3} = \frac{14}{3} = 4\frac{2}{3}$$

10.(B) Put $x^2-1=t$

$$I=\frac{1}{2}\int_0^1 \tan^{-1} t\,dt=\frac{1}{2}\left[\tan^{-1} t.t\,|_0^1-\int_0^1\frac{t}{1+t^2}\,dt\right]$$

$$I=\frac{\pi}{8}-\frac{1}{2}.\frac{1}{2}\ln(2)=\frac{\pi}{8}-\frac{\ln 2}{4}$$

11.(D) $A(a)=2\int_0^{\sqrt{1-a}}\left(\left(1-x^2\right)-a\right)dx=\frac{4}{3}(1-a)^{3/2}$ $\quad\therefore\ A(0)=\frac{4}{3}$



$$A\left(\frac{1}{2}\right)=\frac{4}{3}\left(\frac{1}{2}\right)^{3/2}\Rightarrow\frac{A(0)}{A\left(\frac{1}{2}\right)}=2\sqrt{2}$$

12.(A) $\because\ \int\left(x^9+x^6+x^3\right)\left(2x^6+3x^3+6\right)^{1/3}dx=\int\left(x^8+x^5+x^2\right)\left(2x^9+3x^6+6x^3\right)^{1/3}dx$

Let $2x^9+3x^6+6x^3=t$

$\Rightarrow 18\left(x^8+x^5+x^2\right)dx=dt$ $\quad\therefore\ I=\int\frac{t^{1/3}}{18}dt=\frac{1}{18}.\frac{t^{4/3}}{4/3}+C=\frac{1}{24}t^{4/3}+C$

$\therefore\ AB=24\times\frac{4}{3}=32$

13.(A) Use L'Hosptial's rule

$$\lim_{x\to\infty}\frac{\left(\tan^{-1}x\right)\sqrt{x^2+1}}{x}=\frac{\pi}{2}\lim_{x\to\infty}\frac{\sqrt{x^2+1}}{x}=\frac{\pi}{2}$$

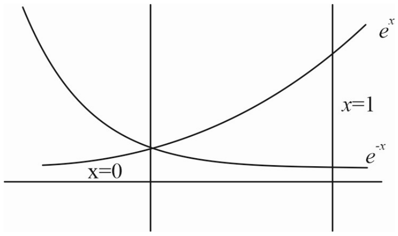
14.(D) $A=\int_0^1\left(x.e^x-\frac{x}{e^x}\right)dx=\frac{2}{e}$

15.(C) $\int_0^{2a} f(x)dx=\int_0^a f(x)dx+\underbrace{\int_a^{2a} f(x)\,dx}_{\substack{Let\,x=2a-t\\ dx=-dt}}=\int_0^a f(x)dx-\int_a^0 f(2a-t)\,dt=\int_0^a f(x)dx+\int_0^a f(2a-x)\,dx=2+4=6$

16.(B) $\int_{-\pi/4}^{\pi/4} a|\sin x|\,dx+b\int_{-\pi/4}^{\pi/4}\frac{\sin x}{1+\cos x}dx+\int_{-\pi/4}^{\pi/4} c\,dx=0$

$\Rightarrow 2a\int_0^{\pi/4}\sin x\,dx+0+c\left(\frac{\pi}{2}\right)=0$ $\quad\Rightarrow 2a\left(1-\frac{1}{\sqrt{2}}\right)+0+\frac{\pi c}{2}=0$ $\quad$ (by property)

17.(D) Area $\int_0^1 \left(e^x - e^{-x}\right)dx = \left(e^x + e^{-x}\right)_0^1$



$= e + \frac{1}{e} - 2$

18.(C) One of its period is 3.

So that $\int_{15}^{45} f(x)dx = 10\int_0^3 f(x)dx = 50$

19.(B) $g(x) = \int(\pi\sin\pi x + 2x - 4)dx = -\cos\pi x + x^2 - 4x + 5(\because g(1) = 3) = (x-2)^2 + (1-\cos\pi x)$

20.(D) $I = \int_1^{2013} (x-1)(x-2)(x-3).....(x-2013)dx$

Using $\int_1^{2013} (2013-x)(2012-x)......(1-x)dx = -I \qquad \Rightarrow 2I = 0 \therefore I = 0$

SECTION-2

21.(1) $I = \int_{-1}^{1} xdx - \int_{-1}^{1}[2x]dx = \left(\frac{x^2}{2}\right)_{-1}^{1} - \left\{\int_{-1}^{-1/2}[2x]\,dx + \int_{-1/2}^{0}[2x]\,dx + \int_{0}^{+1/2}[2x]\,dx + \int_{1/2}^{1}[2x]\,dx\right\}$

$= 0 - \left\{\int_{-1}^{-1/2}(-2)\,dx + \int_{-1/2}^{0}(-1)\,dx + \int_{0}^{+1/2}0dx + \int_{1/2}^{1}1.dx\right\} = 1$

22.(1.57) $x + \pi = t$

$dx = dt$

$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left[t^3 + \cos^2(2\pi + t)\right]dt$

$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} t^3dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos^2 tdt = 0 + 2\int_0^{\pi/2}\cos^2 tdt$

23.(2) $I=\int_0^{\frac{1}{2}}\frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{1/4}}dx=\int_0^{\frac{1}{2}}\frac{1+\sqrt{3}}{\left((x+1)^8\left(\frac{1-x}{1+x}\right)^6\right)^{1/4}}dx=\int_0^{\frac{1}{2}}\frac{1+\sqrt{3}}{(x+1)^2\left(\frac{1-x}{1+x}\right)^{3/2}}dx$

Put $\frac{1-x}{1+x}=t\Rightarrow\frac{-2dx}{(1+x)^2}=dt \qquad \Rightarrow I=-\frac{1}{2}\int_1^{1/3}\frac{1+\sqrt{3}}{t^{3/2}}=-\frac{1}{2}\times\left(1+\sqrt{3}\right)\left[\frac{t^{-1/2}}{-\frac{1}{2}}\right]_1^{1/3}=\left(1+\sqrt{3}\right)\left(\sqrt{3}-1\right)=2$

24.(9) $\alpha=\int_0^1\left(e^{9x+3\tan^{-1}x}\right)\left(\frac{12+9x^2}{1+x^2}\right)dx=\int_0^1\left(e^{9x+3\tan^{-1}x}\right)\left(9+\frac{3}{1+x^2}\right)dx=\left(e^{9x+3\tan^{-1}x}\right)\Big|_0^1=e^{9+\frac{3\pi}{4}}-1$

$\Rightarrow \ln\left|(1+\alpha)\right|-\frac{3\pi}{4}=9$

25.(8) $I=\int_0^{\pi}\frac{x^2\sin 2x\sin\left(\frac{\pi}{2}\cos x\right)dx}{2x-\pi} \qquad \therefore\ I=\int_0^{\pi}\frac{(\pi-x)^2\sin 2(\pi-x)\sin\left(\frac{\pi}{2}\cos(\pi-x)\right)dx}{2(\pi-x)-\pi}$

$\therefore\ I=\int_0^{\pi}\frac{-(\pi-x)^2\sin 2x\sin\left(-\frac{\pi}{2}\cos x\right)dx}{\pi-2x}$

$\therefore\ I=-\int_0^{\pi}\frac{(\pi-x)^2\sin 2x\sin\left(\frac{\pi}{2}\cos x\right)dx}{2x-\pi} \qquad \therefore 2I=\int_0^{\pi}\frac{\left\{x^2-(\pi-x)^2\right\}\sin 2x\sin\left(\frac{\pi}{2}\cos x\right)dx}{2x-\pi}$

$\therefore 2I=\int_0^{\pi}\pi 2\sin x\cos x\sin\left(\frac{\pi}{2}\cos x\right)dx \qquad \therefore\ Let\frac{\pi}{2}\cos x=t\Rightarrow-\frac{\pi}{2}\sin xdx=dt$

$\therefore 2I=-\int_{\pi/2}^{-\pi/2}8\frac{t}{\pi}\sin tdt=\frac{16}{\pi}\left[-t\cos t+\sin t\right]_0^{\pi/2}=\frac{16}{\pi} \qquad \Rightarrow I=\frac{8}{\pi}\Rightarrow A=8$